

Project Design Phase

Solution Architecture

Date	24 June 2026
Name	M Manosika
Project Name	Automated Network Request Management System
Maximum Marks	4 Marks

Solution Architecture Description

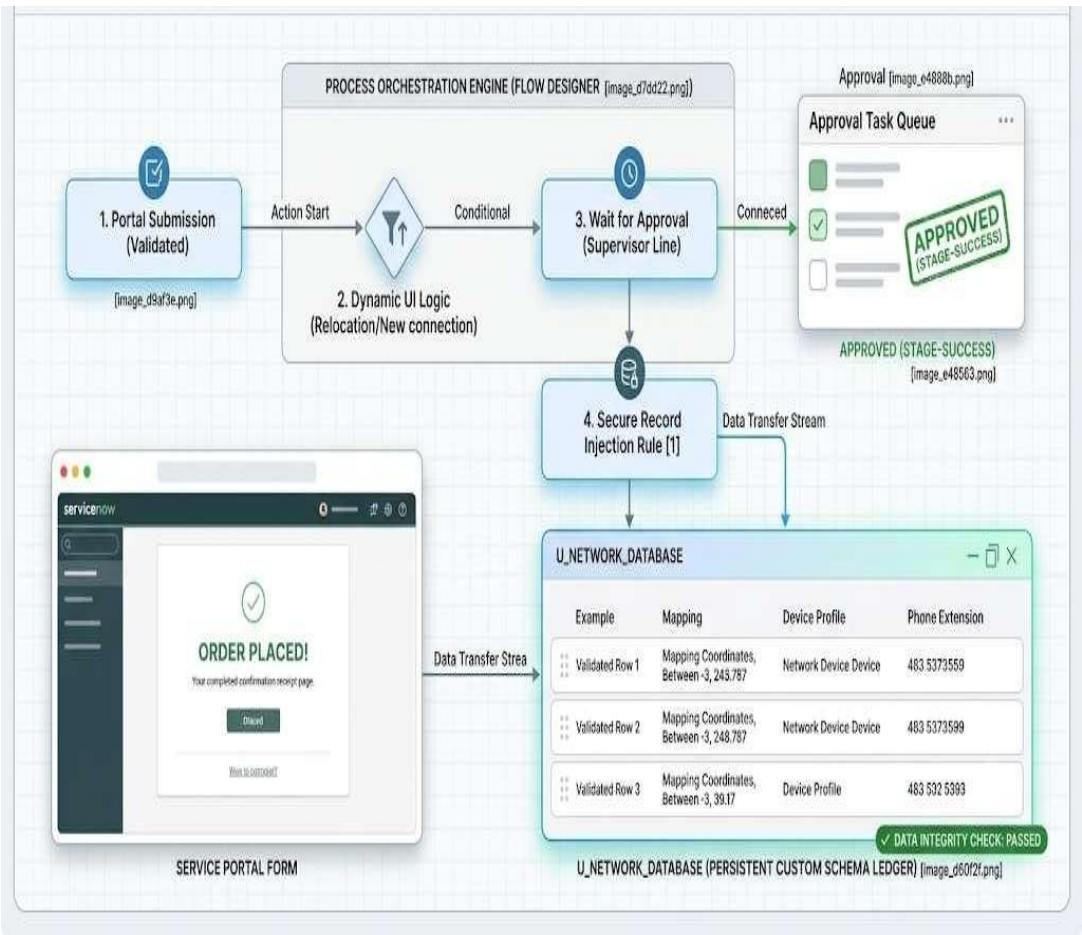
The Solution Architecture defines the overall structure of the Automated Network Request Management System and explains how different ServiceNow components work together to automate the complete request lifecycle. It provides a secure, organized, and scalable framework that manages network requests from submission to final completion.

The primary objectives of the proposed architecture are:

- **Improve Business Operations:** Replace manual request handling, email-based approvals, and repetitive data entry with an automated workflow.
- **Provide Process Transparency:** Clearly define the interaction between users, approval workflows, business logic, and database storage.
- **Support System Design:** Describe the communication between the Service Portal, Flow Designer, and backend database throughout the request lifecycle.
- **Maintain System Reliability:** Ensure secure data processing through validation rules, access control mechanisms, and automated database updates.

Solution Architecture & Data Flow

The solution follows a layered architecture where network requests move through multiple processing stages before being securely stored in the database. Each stage validates the request and ensures accurate execution of business processes.



Network request Management

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Item	Requested for	Quantity	Price
Network Request	 Abraham Lincoln	1	\$90.00

Stage

▶ Assess or Scope Task

Activity	Attachments	Additional Details
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Architecture Workflow

1. Request Submission Layer (Service Portal)

The workflow begins when an employee accesses the **Network Request Catalog** through the ServiceNow Service Portal. During request creation, the system automatically retrieves employee information and validates mandatory fields. UI Policies dynamically display only the fields relevant to the selected request type, making the form simple and user-friendly.

2. Workflow Processing Layer (Flow Designer)

After the request is submitted, it is transferred to **ServiceNow Flow Designer**, which acts as the central workflow engine. This layer validates the request, executes business rules, and controls the overall processing sequence without requiring manual intervention.

3. Approval Management Layer

The workflow automatically forwards the request to the reporting manager for approval. Notifications are generated through the ServiceNow platform to keep both the requester and the approver informed. Only approved requests are allowed to proceed to the next stage, ensuring compliance with organizational policies.

4. Data Storage Layer

Once approval is received, the validated request information is securely recorded in the **u_network_database** table. This automated process

eliminates manual data entry, maintains data accuracy, and creates a permanent record for future tracking, reporting, and auditing. Finally, the requester receives a confirmation that the network request has been successfully processed.